Course Project 2

Group 16

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a. Dataset name

We have used the **thermal comfort dataset** and we have analyzed the dataset for CO2 levels, Illuminance, Occupant count and Temperatture for the Room 1 available in the datatset

- co2_room_1.csv
- Illuminance_room_1.csv
- occupant_count_room_1.csv
- temperature_room_1.csv

b. Objectives for the clustering task

1. **Identify typical room usage modes** (unoccupied, lightly-occupied, fully-occupied) by clustering combined occupancy / CO_2 / temperature / illuminance patterns.

2. **Detect ventilation / comfort issues**: clusters with high CO_2 and rising temperature despite occupancy indicate insufficient ventilation or wrong HVAC scheduling.

3. **Provide actionable recommendations** (increase ventilation, change setpoints, schedule changes, sensor maintenance) based on cluster properties.

c. Approach & framework

To perform clustering on the room dataset, we first synchronized and merged all the time-series files (CO_2, illuminance, occupancy, and temperature) to a common one-minute interval, interpolating small gaps where necessary. This ensured that all variables were aligned and comparable. The key features chosen for clustering were occupancy, CO_2 concentration, temperature, and illuminance, since they directly reflect both human activity and indoor environmental conditions. In addition, we derived cyclic features such as hour-of-day (using sine and cosine transformations) to capture daily usage patterns and HVAC scheduling effects. These features were standardized using z-score scaling to eliminate bias due to different measurement units. For clustering, we applied the KMeans algorithm, selecting the optimal number of clusters based on silhouette scores to balance cohesion and separation. Finally, the resulting clusters were analyzed both numerically (cluster centers) and visually (PCA projection and time-series overlays) to interpret system behavior.

d. Observations from typical clusters & practical relevance (how to interpret)

The clustering results revealed distinct operating modes of the room environment. One cluster represented unoccupied periods, characterized by low CO_2 levels, low illuminance, and stable temperatures—indicating that the HVAC could run at setback or minimal mode during these times. Another cluster captured normal occupied conditions, where moderate CO_2 and temperature levels suggested that ventilation and cooling were largely adequate to maintain comfort. A third cluster corresponded to high occupancy with elevated CO_2 concentrations and rising temperatures, pointing to insufficient ventilation or delayed VAV response under heavier loads. Occasional outlier clusters reflected anomalies such as sudden CO_2 spikes or abrupt changes in lighting, which may be due to sensor errors or unusual events. These observations are practically relevant as they highlight opportunities for energy savings during unoccupied hours, the need to optimize ventilation for dense occupancy periods, and the value of anomaly detection for sensor maintenance and system tuning.